SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA0301
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

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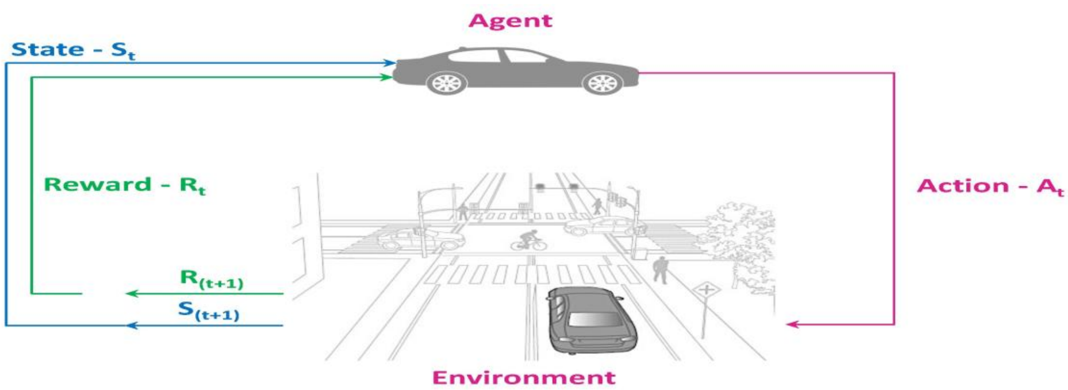
QUESTIONS: 1 **Total Marks: 50**

Annexure A
Analytical Problem-Solving Assessment

1.Q-Learning vs Deep Q-Network (DQN) for Autonomous Vehicles



1. Introduction
An autonomous vehicle must continuously make decisions such as accelerating, braking, changing lanes, and maintaining a safe distance from other vehicles. Reinforcement Learning (RL) can be used to train the vehicle to make these decisions based on its observations of the driving environment.
Two important RL approaches are **Q-Learning** and **Deep Q-Network (DQN)**. Both methods learn the value of taking an action in a particular state, but they differ significantly in how they represent and learn these values.

2. Q-Learning
Q-Learning is a model-free, value-based Reinforcement Learning algorithm. It maintains a **Q-table** containing the expected value of taking an action in a particular state.
For an autonomous vehicle, the states can include:

- Vehicle speed

- Distance from the vehicle ahead
- Traffic-light status
- Lane position
- Distance from obstacles
- Road condition

Possible actions can be:

- Accelerate
- Brake
- Maintain speed
- Change lane left
- Change lane right

The Q-value is updated using the current reward and the estimated future value.

The general Q-Learning process is:

Observe State → Select Action → Receive Reward → Update Q-Value → Select Better Action

For example, if the vehicle safely maintains distance from another vehicle, it receives a positive reward. If it gets too close to an obstacle, it receives a negative reward.

3. Deep Q-Network (DQN)

DQN improves Q-Learning by replacing the Q-table with a **Deep Neural Network**.

Instead of storing a separate Q-value for every possible state, the neural network receives the current state as input and predicts Q-values for possible actions.

For example:

Input:

- Vehicle speed
- Distance to obstacle
- Lane position
- Traffic density
- Traffic-light information

Neural Network → Q-values

Output:

- Accelerate = 2.5
- Brake = 8.7
- Maintain speed = 5.2
- Change lane = 6.4

The vehicle selects the action with the highest suitable Q-value while still using exploration during training.

4. Architecture

A simple DQN-based autonomous vehicle system can be represented as:

Autonomous Vehicle

↓

Camera / Sensors

↓

Current State

↓

Speed + Lane + Obstacles + Traffic Signal

↓

Deep Neural Network (DQN)

↓

Q-Values
↓
Accelerate / Brake / Maintain / Left / Right
↓
Selected Action
↓
Vehicle Movement
↓
Reward
↓
DQN Learning
↺

5. Learning Performance Comparison
Q-Learning
Q-Learning performs well when the number of states and actions is small.
However, autonomous driving has a very large state space. For example, different vehicle speeds, positions, traffic conditions, obstacles, and road situations can create millions of possible states.
Maintaining a Q-table for all these states becomes inefficient.
Therefore, Q-Learning has limited scalability for complex autonomous driving environments.
DQN
DQN can handle much larger and more complex state spaces because a neural network approximates the Q-function.
Instead of storing every possible state in a table, the network learns patterns from the observed states.
Therefore, DQN is more suitable for complex driving environments.

6. Convergence Speed
Q-Learning can converge relatively quickly in small and simple environments because updating a Q-table is computationally simple.
However, in a complex driving environment, it may require a very large number of state-action combinations to be explored. This makes learning slow and inefficient.
DQN requires more computation for neural-network training. Initial learning may therefore be computationally expensive.
However, once the neural network learns useful representations, DQN can generalize across similar states. This makes it more practical for large environments.
Therefore:
Simple environment → Q-Learning may converge faster
Complex environment → DQN provides better scalability and generalization

7. Scalability
Scalability is one of the biggest differences between Q-Learning and DQN.

Feature	**Q-Learning**	**DQN**
Value representation	Q-table	Neural network
Small state space	Very suitable	Suitable
Large state space	Poor scalability	Good scalability
Complex environments	Limited	Highly suitable

Feature	Q-Learning	DQN
Memory requirement	Can become very large	Neural-network parameters
Generalization	Low	High
Training computation	Lower	Higher
Autonomous driving	Limited	More suitable

8. Why DQN is Better for Complex Driving

Autonomous vehicles operate in environments with many possible situations.
For example, the vehicle may encounter:

- Heavy traffic
- Pedestrians
- Sudden braking
- Obstacles
- Different road layouts
- Traffic signals
- Multiple lanes
- Changing weather conditions

A Q-table cannot efficiently represent all these combinations.
DQN uses a neural network to approximate the Q-function and can learn relationships between similar situations. Therefore, it can generalize better to previously unseen but related states.
DQN also commonly uses techniques such as **experience replay** and a **target network** to make training more stable.

9. Overall Analysis

Q-Learning is simple, easy to understand, and effective for environments with a limited number of states and actions. However, its Q-table becomes impractical when the state space becomes very large.
DQN solves this limitation by using a deep neural network to approximate Q-values. Although DQN requires more computational resources and careful training, it is much more scalable for complex autonomous driving environments.
For example, a small simulated vehicle moving on a simple grid can use Q-Learning effectively. A realistic autonomous vehicle dealing with traffic, obstacles, pedestrians, and different road conditions is better suited to DQN.

10. Conclusion

For autonomous vehicle applications, **DQN is generally more suitable than traditional Q-Learning** because autonomous driving has a large and complex state space.
Q-Learning is simpler and may learn quickly in small environments, but its scalability is limited by the size of the Q-table. DQN requires greater computational power and training time, but it provides better scalability, generalization, and learning capability for complex driving environments.
Therefore, the comparison can be summarized as:
Q-Learning → Simple, low computational complexity, suitable for small environments
DQN → More computationally demanding, but scalable and suitable for complex autonomous driving
Hence, **DQN provides better overall performance and scalability for realistic autonomous vehicle environments.**

Code of Conduct:

I **<u>Yoga Madha A-192425058</u>** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yogamadha